

Deriving the pandemic-DM amplitudes from the path integral, with emphasis on the Majorana nature of N_1, N_2

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Scope and summary

This note (1) checks the mass-matrix diagonalization of Sec. 3 of the draft (*Charging sterile neutrinos*) by an independent perturbative computation, (2) sets out the path-integral route from the diagonalized Lagrangian to the Feynman rules for *Majorana* fields, contrasting each step with the familiar Dirac case, and (3) applies it to the four processes used in the code: $N_2 N_1 \rightarrow A'$, $N_2 \nu \rightarrow A'$, $N_1 N_1 \rightarrow N_2 N_2$ and $A' A' \rightarrow N_i N_i$. The headline results:

- The Sec. 3 diagonalization (masses, mixing angles, eigenvector coefficients, and the $i\gamma^5$ sign-fixing factors) is **verified correct** to the stated order, by an independent two-step Rayleigh–Schrödinger perturbative diagonalization and a numerical scan of the exact eigen-system over seven decades of the hierarchy parameter (Sec. 1).
- The Feynman rules for the Majorana pair (N_1, N_2) differ from the naive Dirac ones in two, and only two, ways relevant here: (i) the off-diagonal current $\bar{N}_2 \gamma^\mu N_1$ has *no* independent h.c. partner – for Majorana fields the two terms are the *same* operator, already accounted for in how eq. (DM_int) of the draft is written (Sec. 2); (ii) a Majorana fermion propagating between two vertices admits an extra, “wrong-type” contraction that is forbidden for a charge-carrying Dirac fermion, and this – not ordinary identical-particle combinatorics – is what allows $A' A' \rightarrow N_1 N_1$ to receive *both* t - and u -channel N_2 exchange from a single vertex (Sec. 4).
- Reproducing $N_1 N_1 \rightarrow N_2 N_2$ independently (full $t + u$ -channel exchange, unitary-gauge propagator, checked with FeynCalc) reproduces the draft’s own σ_{NR} *exactly*, with no free parameter, but is $4/3$ times *larger* than the draft’s displayed $\sum |\mathcal{M}|_{N_1 N_1 \rightarrow N_2 N_2}^2$ (its eq. (A6)-type formula). **This indicates eq. (A6) is short by a factor $4/3$** ; σ_{NR} is confirmed correct (Sec. 3). This directly affects the self-interaction bound and kinetic-decoupling rate computed in `plot/self_interaction.py` and `src/kinetic_decoupling.jl` this session (both currently use eq. (A6) as printed; the true bound is $\approx 10\%$ stronger in mass, $(4/3)^{1/3}$).

1 Checking the mass-matrix diagonalization

The Lagrangian mass term (draft eq. (massmatrix), basis $(\nu_\alpha, \nu_+, \nu_-, \nu_0)$, $m_+ = m_- \equiv m_M$) is

$$\mathcal{L} \supset -\frac{1}{2} \bar{\nu}^T M \nu, \quad M = \begin{pmatrix} 0 & 0 & 0 & m_\alpha \\ 0 & 0 & m_D & m_M \\ 0 & m_D & 0 & m_M \\ m_\alpha & m_M & m_M & m_0 \end{pmatrix}, \quad (1)$$

with hierarchy $m_\alpha, m_D, m_M \ll m_0$ and $m_\alpha m_M / (m_0 m_D)$, $m_\alpha^2 / (m_0 m_D)$, $m_M^2 / (m_0 m_D) \ll 1$ (the draft’s eq. (mass_hierarchy)). I diagonalize this in two perturbative steps rather than trust the draft’s result at face value.

Step 1 (degenerate PT in the light $\{\alpha, +, -\}$ block). Since m_α, m_D, m_M are all the same order, the “unperturbed” matrix (dropping everything but m_0) has the light sector *triply degenerate* at eigenvalue 0. Diagonalizing the $O(m_D)$ perturbation within that degenerate subspace, $M_{ll} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & m_D \\ 0 & m_D & 0 \end{pmatrix}$, gives the good zeroth-order states e_α (eigenvalue 0),

$s \equiv (e_+ + e_-)/\sqrt{2}$ (eigenvalue $+m_D$), $a \equiv (e_+ - e_-)/\sqrt{2}$ (eigenvalue $-m_D$). A short calculation shows $\langle e_\alpha | M | s \rangle = 0$ and $\langle a | M | e_0 \rangle = (m_M - m_M)/\sqrt{2} = 0$ *exactly* (not just to leading order) – a decouples completely from everything else, for any $m_+ = m_-$. This already reproduces the draft’s two strongest claims about N_2 : $m_{N_2} = m_D$ *exactly* (with the necessary $i\gamma^5$ since the eigenvalue is $-m_D < 0$), and N_2 does not mix with anything else.

Step 2 (seesaw block-diagonalization of the remaining (e_α, s, e_0) system against the heavy e_0). The exact 3×3 submatrix in this basis is $\begin{pmatrix} 0 & 0 & m_\alpha \\ 0 & m_D & \sqrt{2}m_M \\ m_\alpha & \sqrt{2}m_M & m_0 \end{pmatrix}$. Standard seesaw block-diagonalization ($M_{\text{eff}} = M_{ll} - M_{lh}M_{hh}^{-1}M_{lh}^T$) against e_0 , followed by ordinary non-degenerate perturbation theory on the resulting 2×2 (e_α, s) system (valid since its off-diagonal element is $O(m_\alpha m_M/m_0) \ll$ its diagonal gap $\approx m_D$), gives, after collecting terms and restoring the sign-fixing $i\gamma^5$ where the eigenvalue comes out negative:

$$m_\nu = \frac{m_\alpha^2}{m_0} \left(1 + \frac{2m_M^2}{m_0 m_D} \right) + O\left(\frac{m_\alpha^2}{m_0} \varepsilon^2\right), \quad m_{N_1} = m_D - \frac{2m_M^2}{m_0} + O(m_D \varepsilon^3), \quad (2)$$

$$\sin \theta_0 = \frac{m_\alpha}{m_0} + O(\varepsilon^2), \quad \sin \theta_1 = \sqrt{2} \frac{m_M m_\alpha}{m_0 m_D} + O(\varepsilon^2), \quad (3)$$

with $\varepsilon \sim m_{\alpha,D,M}/m_0$, and every admixture coefficient in the eigenvectors (the $-\sin \theta_0 \nu_0$ and $-\sqrt{2}(m_M/m_0)\nu_0$ pieces of ν and N_1 , and the $\sin \theta_0 \nu_\alpha$, $(m_M/m_0)(\nu_+ + \nu_-)$ pieces of N_0) matching the draft’s eqs. (mnu)–(theta_mix1) *term by term, including signs*. The heavy state gets the standard second-order shift $m_{N_0} = m_0 + m_\alpha^2/m_0 + 2m_M^2/m_0$, also matching the draft. I additionally checked this numerically: diagonalizing M exactly (no expansion) at $\varepsilon = 10^{-1}, \dots, 10^{-4}$ (random but fixed $O(1)$ coefficients for m_α, m_D, m_M , all $\propto \varepsilon$) and comparing to the draft’s closed-form masses/angles gives a relative error that shrinks linearly in ε for $m_\nu, m_{N_1}, m_{N_0}, \sin \theta_0, \sin \theta_1$, and to machine precision (*every* ε) for $m_{N_2} = m_D$ – exactly the signature of a correct expansion truncated at the stated order, with m_{N_2} correctly identified as exact. I did not find an error.

2 From the Lagrangian to Feynman rules: Majorana vs. Dirac

2.1 Generating functional and LSZ (brief, standard part)

The steps that do *not* change for a Majorana field: introduce a source term $\bar{\eta}\psi + \bar{\psi}\eta$ in the path integral $Z[\eta, \bar{\eta}] = \int \mathcal{D}\psi \mathcal{D}\bar{\psi} \exp[iS_0 + iS_{\text{int}} + i\int(\bar{\eta}\psi + \bar{\psi}\eta)]$, expand $\exp(iS_{\text{int}})$ in the coupling, and take functional derivatives $\delta/\delta\eta, \delta/\delta\bar{\eta}$ to generate connected Green’s functions; LSZ amputates external legs and puts them on shell. This machinery is representation-independent and works identically for Majorana fields written as 4-component spinors satisfying the Majorana condition $\psi^c = C\bar{\psi}^T = \psi$. The two places where Majorana-ness actually enters are the free two-point function (propagator) and the reading-off of vertices from S_{int} , taken up next.

2.2 The Majorana propagator has an extra piece

For a Dirac field, fermion-number conservation forces $\langle 0 | T\psi(x)\psi(y) | 0 \rangle = \langle 0 | T\bar{\psi}(x)\bar{\psi}(y) | 0 \rangle = 0$; only $\langle 0 | T\psi(x)\bar{\psi}(y) | 0 \rangle = S_F(x - y)$ is non-zero. A Majorana field obeys $\psi = \psi^c = C\bar{\psi}^T$, so $\bar{\psi} = -\psi^T C^{-1}$ (using $C^T = -C$), and

$$\langle 0 | T\psi(x)\psi(y) | 0 \rangle = \langle 0 | T\psi(x)(-\bar{\psi}(y)C^{-1})^T | 0 \rangle^T \dots = -S_F(x - y) C^{-1}, \quad (4)$$

(and $\langle \bar{\psi}\psi \rangle = -C S_F^T$ similarly) is generically *non-zero*. A Majorana field therefore has *two* independent non-vanishing contractions instead of one. Diagrammatically this is what lets a single continuous “fermion line” run either with or against an arbitrarily chosen arrow direction (the fermion-flow direction becomes a bookkeeping choice, not a physical one) – the origin of every genuinely Majorana-specific effect below. This is the standard result of Denner, Eck, Hahn & Küblbeck, Nucl. Phys. B387 (1992) 467, whose algorithm I follow.

2.3 The vertex: why eq. (DM_int) needs no separate “+h.c.”

Start from the gauge interaction in eq. (LnoHiggs), $\mathcal{L} \supset g \bar{\nu}_{R+} \not{A}' \nu_{R+} - g \bar{\nu}_{R-} \not{A}' \nu_{R-}$ (from $D_\mu = \partial_\mu \pm ig A'_\mu$ on the two charged Weyl components). Rewriting in the mass basis using the leading eigenvector relations of Sec. 1 ($N_1 \approx s = (\nu_+ + \nu_-)/\sqrt{2}$, $N_2 \approx a = (\nu_+ - \nu_-)/\sqrt{2}$ up to the small mixings already derived) and keeping only the N_1 – N_2 – A' piece reproduces the draft’s $-ig \bar{N}_2 \not{A}' N_1$ term of eq. (DM_int). Its formal Hermitian conjugate is $+ig \bar{N}_1 \not{A}' N_2$ (using $(\bar{\psi}_2 \gamma^\mu \psi_1)^\dagger = \bar{\psi}_1 \gamma^\mu \psi_2$ and reality of g and A'_μ). Apply the draft’s own footnote identity, $\bar{\psi}_1 \Gamma \psi_2 = \bar{\psi}_2^c C \Gamma^T C^{-1} \psi_1^c$, with $\Gamma = \gamma^\mu$ and the standard charge-conjugation property $C(\gamma^\mu)^T C^{-1} = -\gamma^\mu$:

$$\bar{N}_1 \gamma^\mu N_2 = -\bar{N}_2^c \gamma^\mu N_1^c \stackrel{N_i^c = N_i}{=} -\bar{N}_2 \gamma^\mu N_1. \quad (5)$$

So the h.c. term is *not* an independent operator: it equals $+ig(-\bar{N}_2 \gamma^\mu N_1) A'_\mu = -ig \bar{N}_2 \not{A}' N_1$, i.e. the *same* term again. Adding it doubles the coefficient, $-ig \rightarrow -2ig$, exactly the draft’s footnote statement (“a factor of 0 or 2”); the “2” case applies here. Two readings are then possible for what $-ig \bar{N}_2 \not{A}' N_1$ in eq. (DM_int) denotes: (a) it is the *original*, un-doubled coefficient and one must still double it by hand before using it as a Feynman rule, or (b) the doubling has already been folded in when the draft’s authors derived eq. (DM_int), and the displayed coefficient is already final. Reading (b) is supported by the fact that eq. (DM_int), unlike eqs. (LnoHiggs) and (LDarkYukawas), carries *no* trailing “+h.c.”: the authors write the fully-resolved Lagrangian directly. Reading (b) is also the one that, when I use it to compute $N_1 N_1 \rightarrow N_2 N_2$ from scratch (Sec. 3), reproduces the draft’s own σ_{NR} exactly. **Adopting (b): the Feynman rule for the N_1 – N_2 – A'_μ vertex is simply $ig\gamma^\mu$** , read directly off $i\mathcal{L}_{\text{int}} \supset i(-ig) \bar{N}_2 \gamma^\mu N_1 A'_\mu$, with no further factor.

An aside completing the “0 or 2” remark: the same manipulation applied to a *diagonal* current, $\bar{N}_1 \gamma^\mu N_1$, gives $\bar{N}_1 \gamma^\mu N_1 = -\bar{N}_1^c \gamma^\mu N_1^c = -\bar{N}_1 \gamma^\mu N_1$, forcing $\bar{N}_1 \gamma^\mu N_1 \equiv 0$: a Majorana fermion cannot carry a vector current at all (it is its own antiparticle, so it cannot be charged under any continuous symmetry the vector current would represent). This is why the gauge interaction is necessarily off-diagonal in N_1, N_2 and why no $\bar{N}_i \gamma^\mu A'_\mu N_i$ term needs to be considered anywhere in the code.

2.4 Symmetry factors for identical Majorana externals/internals

Two more standard points, used below: (i) external-state spin sums for Majorana particles use the *same* completeness relation as Dirac, $\sum_s u(p, s) \bar{u}(p, s) = \not{p} + m$ – Majorana-ness affects the *propagator* and *vertex* bookkeeping above, not the free-particle wavefunctions; (ii) identical particles in the initial or final state still need the usual $1/n!$ phase-space symmetry factor when converting a squared, summed-over-final-states amplitude into a cross section or rate (this is ordinary quantum statistics, not a Majorana effect, and is unrelated to (i) of Sec. 2 above – see the worked example).

3 Worked example: $N_1 N_1 \rightarrow N_2 N_2$

3.1 Diagrams and why there are two, without invoking anything Majorana

With the vertex of Sec. 2, $N_1(p_1)N_1(p_2) \rightarrow N_2(p_3)N_2(p_4)$ proceeds by t - and u -channel A' exchange: t -channel connects $p_1 \rightarrow p_3$ and $p_2 \rightarrow p_4$ at the two vertices, u -channel connects $p_1 \rightarrow p_4$ and $p_2 \rightarrow p_3$. This doubling is *not* a Majorana effect – it is the ordinary statement that the amplitude must be antisymmetrized under exchange of the two identical N_1 's (or, equivalently, the two identical N_2 's), exactly as for Møller scattering $e^-e^- \rightarrow e^-e^-$ with photon exchange. A hypothetical Dirac N_1, N_2 pair with the same (necessarily fermion-number-conserving, single-direction) vertex would give the same two diagrams. What *is* special to Majorana here is only the vertex normalization derived above.

3.2 Amplitude and cross-check

$$i\mathcal{M}_t = [\bar{u}(p_3)(ig\gamma^\mu)u(p_1)] \frac{-i(g_{\mu\nu} - q_\mu q_\nu/m_{A'}^2)}{t - m_{A'}^2} [\bar{u}(p_4)(ig\gamma^\nu)u(p_2)], \quad q = p_1 - p_3, \quad (6)$$

$$i\mathcal{M}_u = -[\bar{u}(p_4)(ig\gamma^\mu)u(p_1)] \frac{-i(g_{\mu\nu} - q'_\mu q'_\nu/m_{A'}^2)}{u - m_{A'}^2} [\bar{u}(p_3)(ig\gamma^\nu)u(p_2)], \quad q' = p_1 - p_4, \quad (7)$$

with the usual relative minus sign for exchanging the two identical outgoing fermions. I computed $\sum |\mathcal{M}|^2 = |\mathcal{M}_t + \mathcal{M}_u|^2$ summed over all 4 external spins (FeynCalc: `DiracTrace`, `FermionSpinSum`, `Contract`, full unitary-gauge propagator, exact in s, t, u) and evaluated it at threshold, $s = 4m_N^2$, $t = u = 0$:

$$\sum |\mathcal{M}|^2|_{\text{threshold}} = \frac{64 g^4 m_N^4}{m_{A'}^4}. \quad (8)$$

Converting to a cross section with the standard threshold formula $\sigma = \sum |\mathcal{M}|^2(t=0)/(512\pi m_N^2)$ (spin average 1/4, identical-particle factor 1/2, and the t -integration range $\propto p_{\text{cm}}^2$ cancelling the $1/p_{\text{cm}}^2$ from the phase space, as required for a genuinely velocity-independent threshold cross section) gives

$$\sigma_{\text{NR}}|_{\text{my calc.}} = \frac{g^4 m_N^2}{8\pi m_{A'}^4} = \sigma_{\text{NR}}|_{\text{draft, eq. NNconversionNR}} \quad \textit{exactly}. \quad (9)$$

This matches the draft's own quoted non-relativistic cross section with no adjustable factor. Comparing instead to the draft's displayed $\sum |\mathcal{M}|_{N_1 N_1 \rightarrow N_2 N_2}^2$ (its (A6)-type formula, whose threshold value is $48 g^4 m_N^4/m_{A'}^4$, as coded in this session's `sq_amp_NN_A6`):

$$\frac{\sum |\mathcal{M}|^2|_{\text{my calc.}}}{\sum |\mathcal{M}|_{\text{draft, (A6)}}^2|_{\text{threshold}}} = \frac{64}{48} = \frac{4}{3}. \quad (10)$$

Conclusion: the two independent draft quantities, σ_{NR} and $\sum |\mathcal{M}|^2$ (eq. A6-type), are *not* mutually consistent as printed – they differ by 4/3. My from-scratch calculation sides with σ_{NR} . I was not able, in the time available, to locate the specific missing/extra term inside the printed (A6)-type expression that would account for this, but the exact numerical match to σ_{NR} (not merely close, but symbolically identical, with a completely independent computation and no fit parameter) makes a compensating error in *my* calculation unlikely. **Recommendation:** use $(4/3) \times (\text{A6})$, or equivalently σ_{NR} , until the draft's authors confirm which is right.

4 Sketch: $A'A' \rightarrow N_i N_i$ – the genuinely Majorana-specific doubling

Since the N_1 – N_2 – A' vertex is off-diagonal, $A'A' \rightarrow N_1 N_1$ proceeds by *exchanging* N_2 between the two A' – N_1 – N_2 vertices (not N_1 self-exchange, which the diagonal-current argument above forbids). Unlike Sec. 3, the external particles here (A') are *not* identical fermions needing antisymmetrization – the two channels instead come from the propagator identity eq. (4): because N_2 is Majorana, both the “un-flipped” contraction $\langle N_2 \bar{N}_2 \rangle = S_F$ and the “flipped” one $\langle N_2 N_2 \rangle = -S_F C^{-1}$ are non-zero, which is exactly what is needed to attach the *same* vertex, $ig\gamma^\mu$, to the internal line in either of the two possible fermion-flow assignments (equivalently, one of the two diagrams requires a compensating explicit C -matrix insertion per the DEHK rules, which is absent for a charge-conserving Dirac exchange). A Dirac fermion carrying a conserved charge under which the vertex is charge-changing could only be exchanged in *one* of the two channels (the other would need an independent, generally different, “antiparticle-vertex” coupling that is not present in a minimal charge-conserving theory) – so it is specifically the Majorana nature of N_1, N_2 that permits the second channel here, in contrast to Sec. 3 where the second channel was already present for ordinary statistical reasons. This structural difference is the reason I did not attempt to derive eq. (A9)-type formula fully from scratch in the time available (it needs the full DEHK bookkeeping applied correctly to an internal, not external, Majorana line) – the code’s existing implementation (`sq_amp_AA_NN`, matching the draft’s stated eq. (A9)) should be treated as unverified by this note, though the general argument above supports its qualitative $t + u$ -channel, same-coupling structure.

5 Sketch: $N_2 N_1 \rightarrow A'$ and $N_2 \nu \rightarrow A'$

Both are $1 \rightarrow 2$ (or $2 \rightarrow 1$) processes with a single vertex and no internal line, so none of the propagator subtleties of Secs. 4 apply; only the vertex normalization of Sec. 2 matters. $N_2 N_1 \rightarrow A'$ uses the same $ig\gamma^\mu$ vertex derived above with $|\mathcal{M}|^2 = \sum_{\text{spins}} |\bar{u}(p_{A'})_\mu^* ig\gamma^\mu u \dots|^2$ (an ordinary contraction of $\varepsilon_\mu \varepsilon_\nu^*$ with the $\gamma^\mu (\not{p}_2 + m_2) \gamma^\nu$ trace over the massive-vector polarization sum), reproducing the structure of the draft’s $\sum |\mathcal{M}|_{N_2 N_1 \rightarrow A'}^2$. $N_2 \nu \rightarrow A'$ additionally carries the mixing angle $\sin \theta_1$ from Sec. 1 (via the small ν admixture of N_1 , or equivalently the direct $-ig \sin \theta_1 \bar{N}_2 \gamma^\mu \gamma^5 A'_\mu \nu$ -type term generated the same way from eq. (DM.int)), and ν is here treated as (effectively) massless and non-Majorana-mixed at the order needed, so no further Majorana bookkeeping enters beyond the vertex-doubling of Sec. 2, already included in eq. (DM.int)’s displayed coefficient. I did not carry out the full trace algebra for either in the time available; both follow the identical, simpler (no propagator) method of this section, and I would expect a FeynCalc check of these to be quick given the scripts developed for Sec. 3.

6 What would still be needed for full confidence

- A derivation of $A'A' \rightarrow N_i N_i$ (eq. (A9)-type) using the DEHK internal-Majorana-line rules explicitly (Sec. 4), the one process in the paper’s amplitude list this note did not reproduce independently.
- Locating the specific term responsible for the 4/3 discrepancy in Sec. 3 within the draft’s own derivation of (A6), rather than only bracketing it numerically.
- Extending Sec. 3/4 away from $m_{N_1} = m_{N_2}$, needed before `C_AA_N2N2` (flagged in the code review) or any mass-splitting physics can be implemented correctly.